

Electric Vehicles (EE4052)

Sessional-I Exam

Date: September 22, 2025

Course Instructor

Dr. Muhammad Naeem

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 2

Roll No

Section

Student Signature

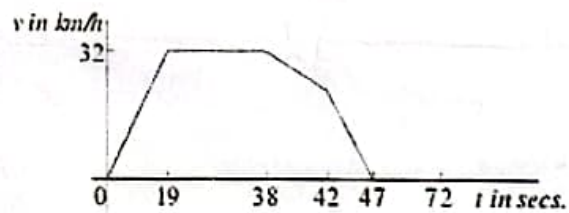
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Attempt all the questions.

CLO 1: Apply the principles of electric vehicle architectures, propulsion systems, and energy storage technologies.

Q1: An EV is tested by standard J227a driving cycle recommended by SAE to analyze the performance of its drivetrain. The driving cycle is shown in figure below and the schedule values are given in table below. Assume no regenerative braking. Also assume level road. [6+4+4+4+2]

Test Parameter	SAE J227a Schedule
Max. Speed, kph	32
Acceleration time, t_a s	19
Cruise time, t_{cr} s	19
Coast time, t_{co} s	4
Brake time, t_b s	5
Idle time, t_i s	25
Total time, s	72
No. of cycles per mile	4



The other parameters of EV are as follows:

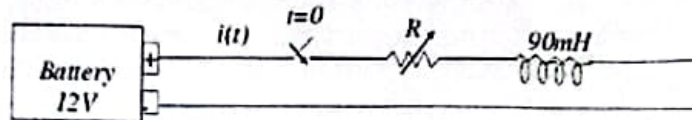
$\rho = 1.16 \text{ kg/m}^3$, $m = 850 \text{ kg}$, $C_D = 0.2$, $A_F = 2 \text{ m}^2$, $g = 9.81 \text{ m/s}^2$, $C_o = 0.009$ and $C_1 = 0$

- Calculate and plot $F_{TR}(t)$ versus t
- Calculate and plot $P_{TR}(t)$ versus t
- Calculate the tractive energy consumption for one cycle
- Calculate the energy loss for one cycle.
- Calculate the energy required for 60 miles of drive based on the energy consumption for one cycle.

$$F_{roll} = mg \cos \beta (C_o + C_1 v^2); \quad F_{AD} = \frac{\rho}{2} C_D A_F v^2; \quad F_{gxT} = mg \sin \beta$$

CLO 2: Analyze the performance and efficiency of electric powertrains, battery systems, and charging infrastructure.

Q2: A 12 V lead-acid battery is connected to a series RL load as shown in figure below. The battery has rated capacity of 80 Ah. At $t = 0$ the switch is closed and the battery begins to discharge. Assume the battery was charged to 80% of its rated capacity at $t = 0$ [2+2+2+2+2]



To analyze the performance of battery:

- Calculate the battery discharge current, $i(t)$, if the constant current discharge rate is $C/5$
- Calculate $SoD(t)$ in Ah
- Calculate $SoC(t)$ in Ah
- Calculate the cut-off time t_{cut} in hours corresponding to 80% DoD
- Calculate the practical available energy in Wh corresponding to 80% DoD